Lecture 1: Big Picture

Usages:

- Positioning & Navigation
- Engineering Geodesy
- Mobile Mapping
- Additonal
  - GNSS Meteorology
  - Time synchronisation
  - Plate tectonics
  - GNSS Reflectometry
  - Reference Systems
  - Orbit determination

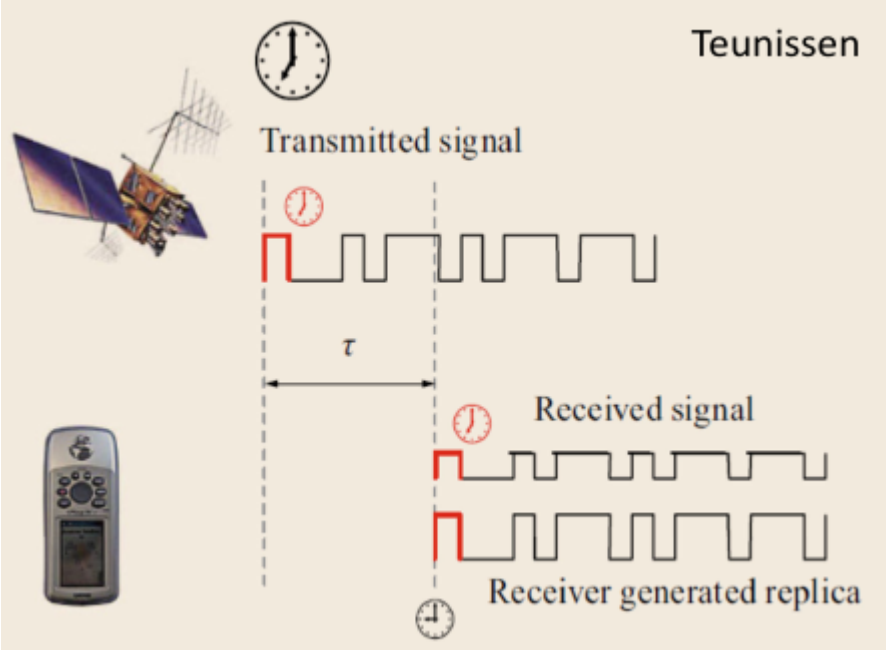
Idea: Measure the distance between the antenna and the several satellites by interpreting the signal

Basic Principle:

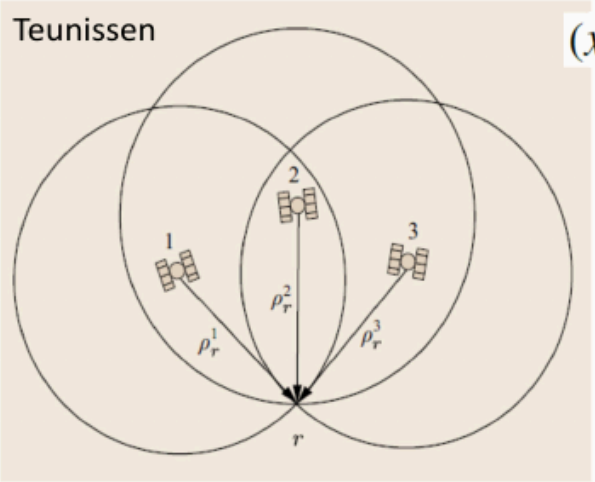
- Satellite coordinates are **known**
- **Measuring distance** between satellite and antenna
- **Calculation** of antenna coordinates
- **Transformation** to useful coordinates

Code Based Ranging (Synchronised Clocks)

1. Satellite transmits signal (**code**) at t
2. Receiver Internal Replica
3. Constant Shifting and Comparison between both signals (**correlation**) until fully overlapping (**Lock**)
4. **Transit Time** τ between t and lock time (Range = $c \cdot \tau$) (~dm-m)
5. Receiver keeps track of the shift (**Tracking**)



$$D = \Delta\tau \cdot c$$



$(x^i, y^i, z^i),\ i = 1, 2, 3$ Satellite positions (known)

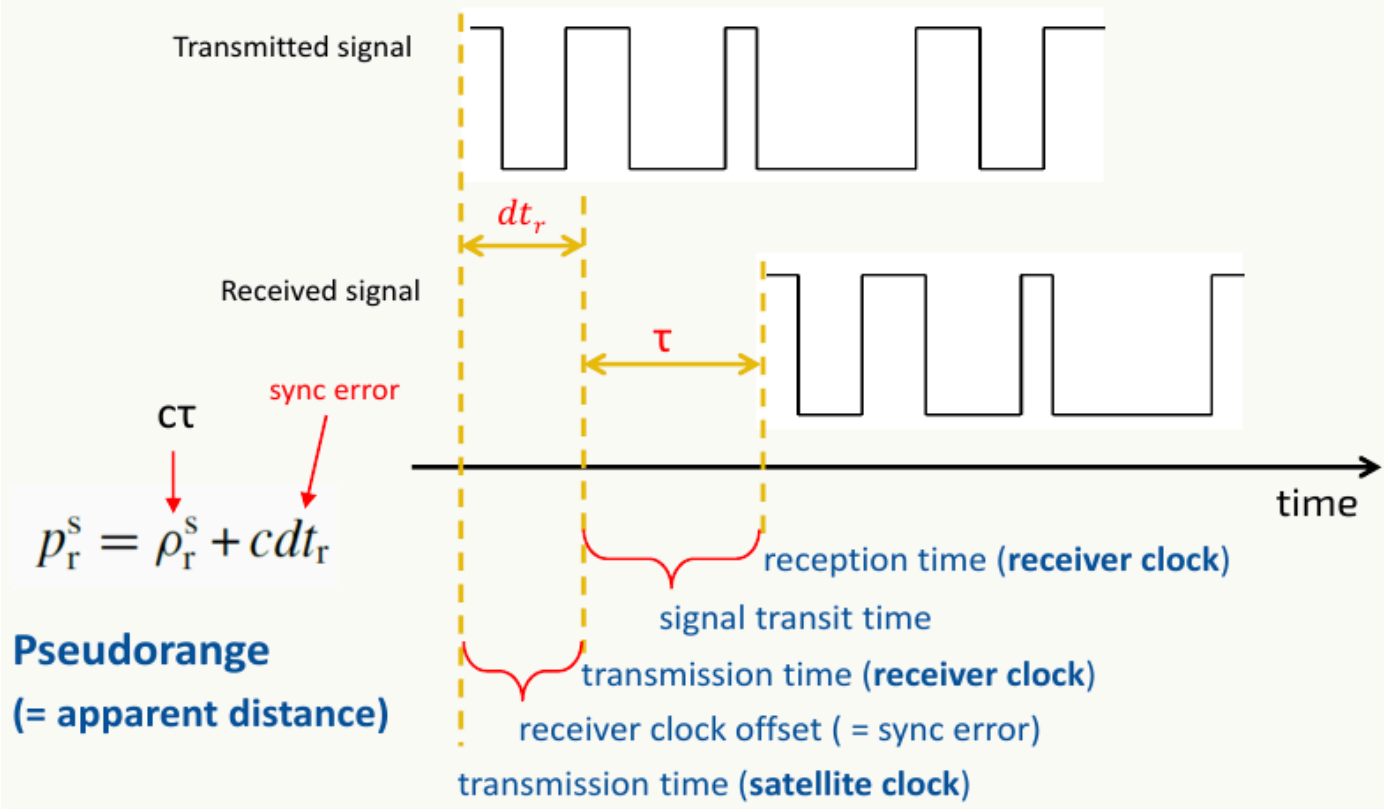
$[\rho_r^1, \rho_r^2, \rho_r^3]$ distance observations (measured) >3

(x_r, y_r, z_r) receiver position (unknown) 3

$$\rho_r^1 = \sqrt{(x_r - x^1)^2 + (y_r - y^1)^2 + (z_r - z^1)^2}$$
$$\rho_r^2 = \sqrt{(x_r - x^2)^2 + (y_r - y^2)^2 + (z_r - z^2)^2}$$
$$\rho_r^3 = \sqrt{(x_r - x^3)^2 + (y_r - y^3)^2 + (z_r - z^3)^2}$$

→ Estimation of (x_r, y_r, z_r)

Code Based Ranging (Non-Synchronised Clocks)



$$(x^i, y^i, z^i), i = 1, 2, 3 \quad \text{satellite positions (known)}$$

$$[p_r^1, p_r^2, p_r^3, p_r^4] \quad \text{pseudoranges (measured)} > 4$$

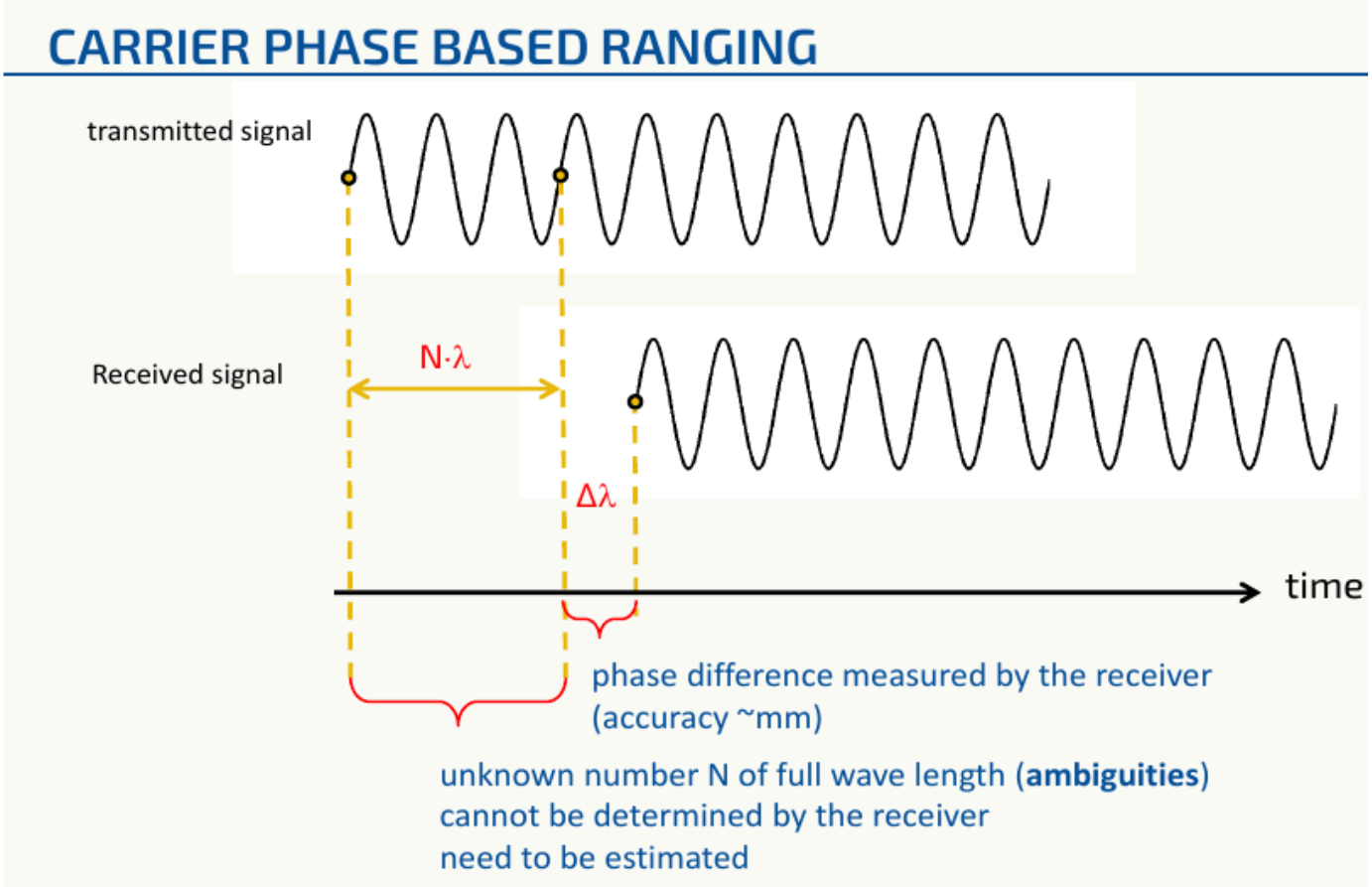
$$\begin{aligned}
p_r^1 &= \sqrt{(x_r - x^1)^2 + (y_r - y^1)^2 + (z_r - z^1)^2} + dt_r \\
p_r^2 &= \sqrt{(x_r - x^2)^2 + (y_r - y^2)^2 + (z_r - z^2)^2} + dt_r \\
p_r^3 &= \sqrt{(x_r - x^3)^2 + (y_r - y^3)^2 + (z_r - z^3)^2} + dt_r \\
p_r^4 &= \sqrt{(x_r - x^4)^2 + (y_r - y^4)^2 + (z_r - z^4)^2} + dt_r
\end{aligned}$$

Estimation of

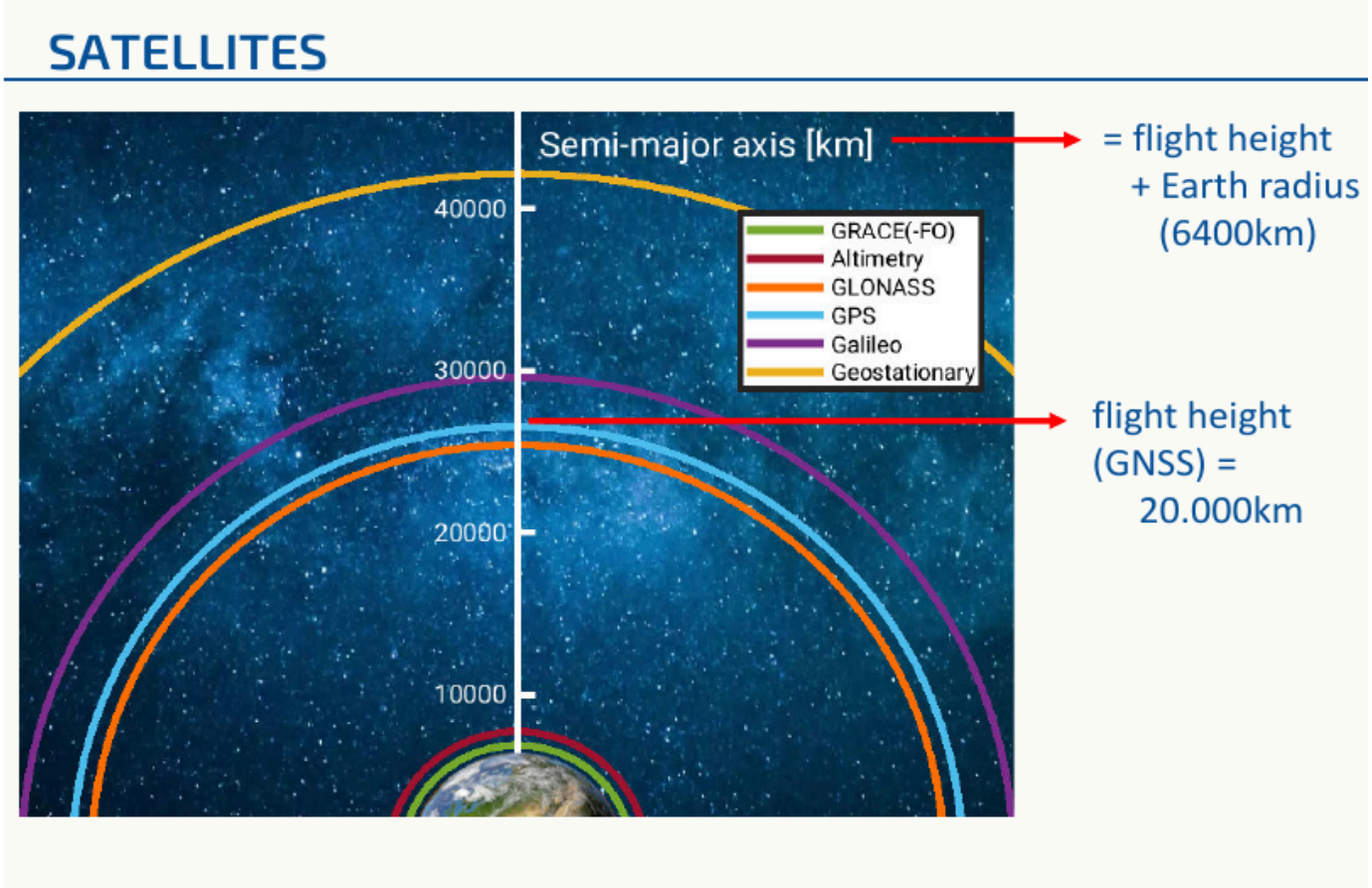
$$\boldsymbol{x} = (x_r, y_r, z_r, dt_r) \quad \text{receiver position} + \text{receiver clock error (unknown)}$$

GNSS Signals

- Code modulation on a carrier wave
- Unique codes for every satellite
- Different Code types C/A and P(Y)
- Different carrier wave frequencies (L1, L2, ...)
- Necessary data (e.g. orbit information) also modulated on the carrier
  - Kepler elements (parametrisation of orbits) are transmitted via the **navigation message**
    - Clock information
    - Time information
    - Position information



- Position and clock offset can be estimated using the known positions of the satellites and the measured transit time
- At **least 4 satellites** due to **4 unknown parameters** (x, y, z, dt)
  - Clock bias is an unknown paramter and needs to be estimated together with the position
- *For cm accuracy , carrier phases and ambiguities for every phase observation*
- High precision GNSS positioning (~cm) needs carrier phase measurements



Reference Systems for Time and Space

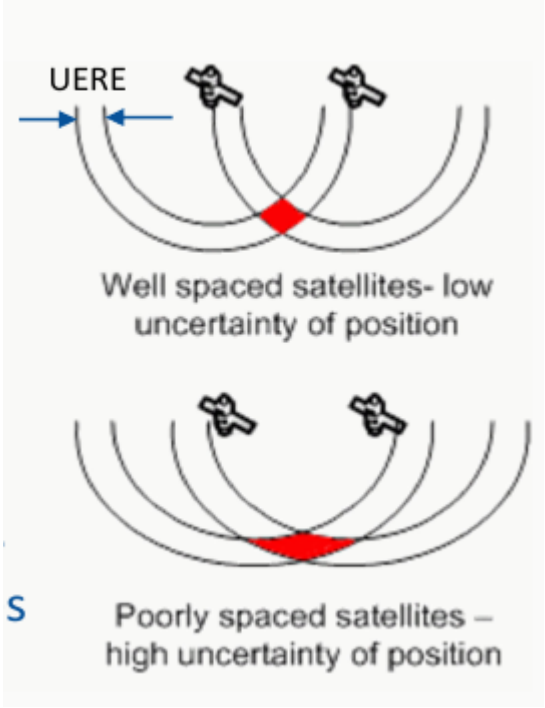
- Reference system: WGS84
  - nearly identical to ITRF
- Transformation to useful coordinates (e.g. ETRS89 UTM with geoid heights)
- Official GPS time is GPST and different to UTC (leap seconds)

In case of a position fix, every receiver is a very accurate UTC clock (nanoseconds accuracy, no drifts

UERE & DOP

All errors/biases are expressed as a range
-> User Equivalent Range Error (**UERE**)

UERE influences position accuracy depending on the number and constellation of satellites
-> **Dilution of Precision**

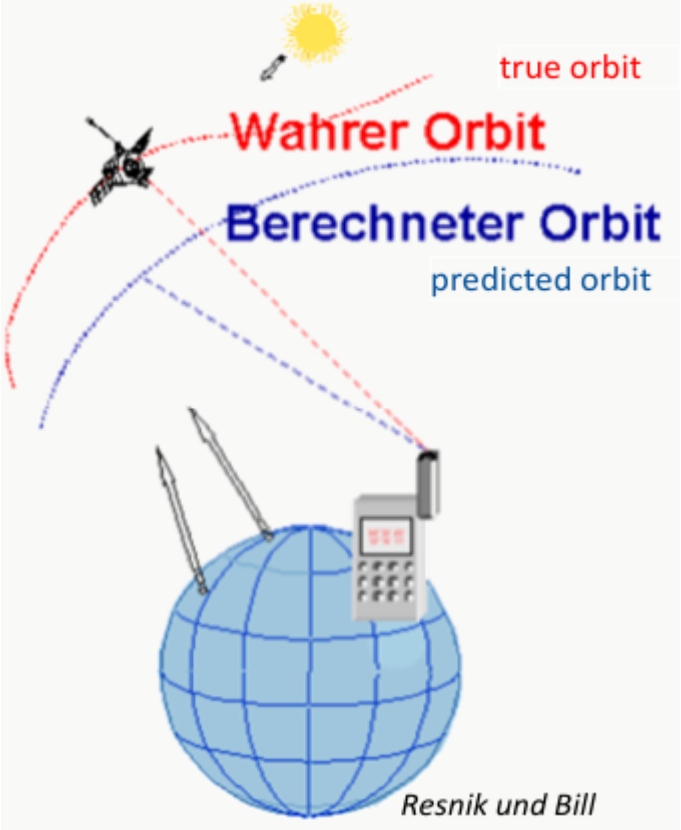


Observation Errors (or biases)

User errors

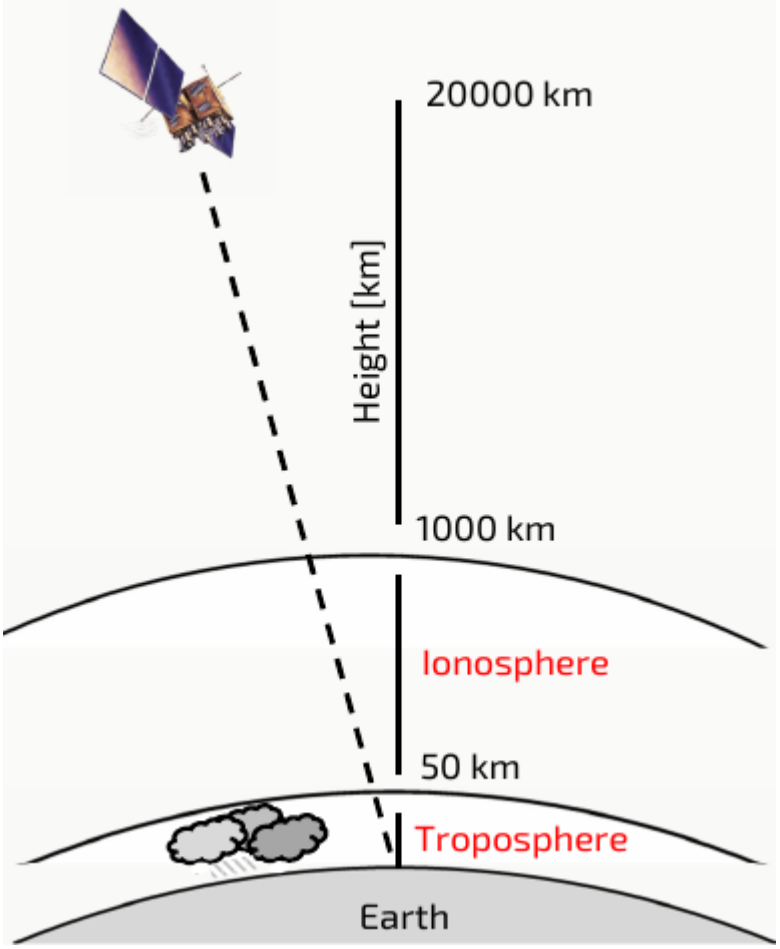
- Antenna
- Receiver (Clock and hardware)
- Reference point in antenna depends on elevation and azimuth of satellite (~mm)
    - can be calibrated
- electronic components introduce **noise** (code ~m, carrier phases ~mm)
    - The only random noise in GNSS

Satellite errors

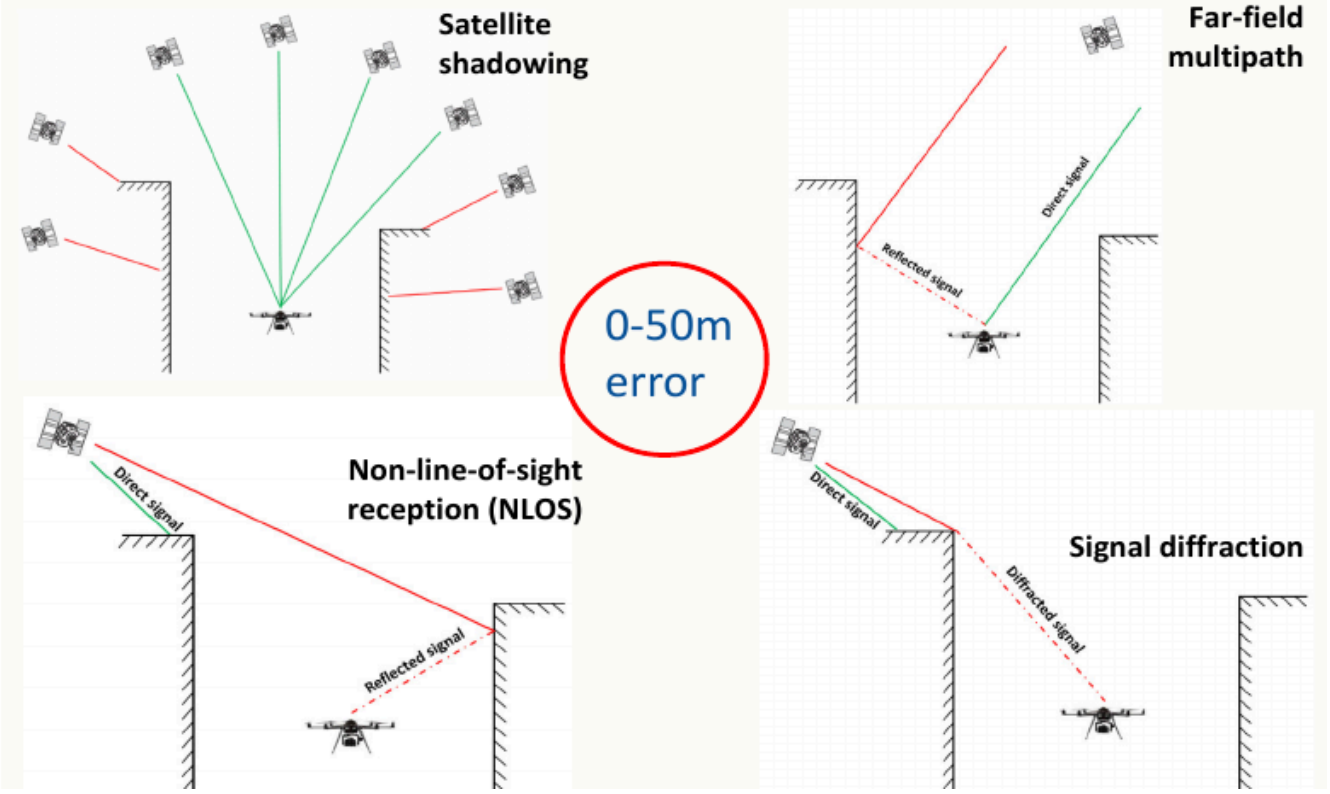


- **Satellite positions**
  - transmitted via navigation messages are not exactly correct (~1m)
- **Satellite Clock error corrections**
  - transmit via navigation message are not exactly correct (~1m)
- Clock and orbit parameters downloadable for postprocessing with an accuracy in **cm**

Signal Path errors



- Above 1000km: Vacuum -> Propagation @ c
- **Ionospheric** Refraction (50-1000km)
  - Ionised gas
  - Delay depends on
    - **free electrons** (day/night, sun activity)
    - **Signal frequency** (**dipersive** medium)
    - -> Multiple frequencies -> ~dm
  - Models - Data from navigation message -> 0-5m
  - Change of speed and direction (**refraction**)
- **Tropospheric** Refraction
  - Electrically neutral
  - Air & Weather -> Varying
  - If unmodelled -> error: 2.4m (90° ev) - 24m (5° ev)
  - **Models** to reduce to 20cm
- Multipath
- ! The **lower** the satellites **elevation angle**, the **longer the way** through the atmosphere, the **higher the effect** of refraction



Reduce errors $\rightarrow$ Relative positioning

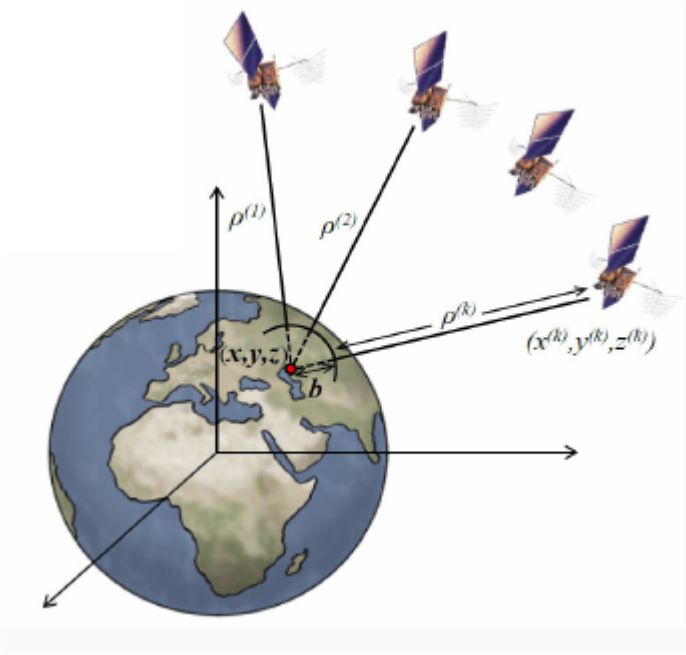
- Signals from same satellite at two receivers are affected by the same errors
- **Master/base** station (known position) and **rover**
- Building differences of observations have no/reduced errors
- Provides vector between receivers (**Baseline vector**)
- One-way communication for **real-time** applications

Positioning/Processing Modes

Single Point Positioning

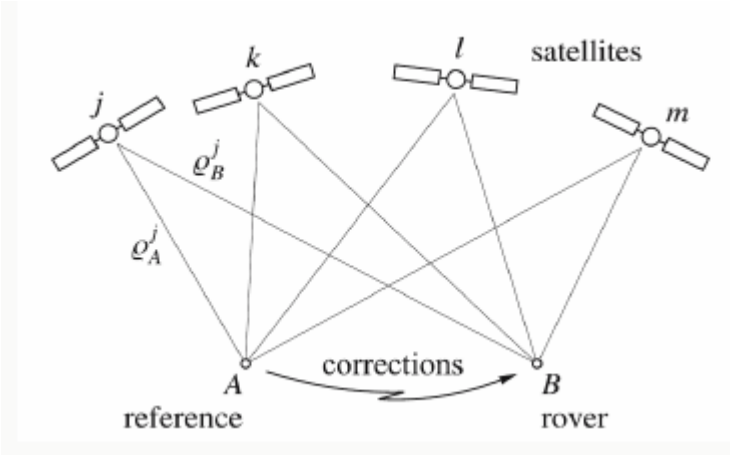
- Observations: code
- Accuracy: 3-20m, absolute
- Real-Time: yes
- Used in mobile phones and navigation devices

- very cheap receivers



Differential GPS (DGPS)

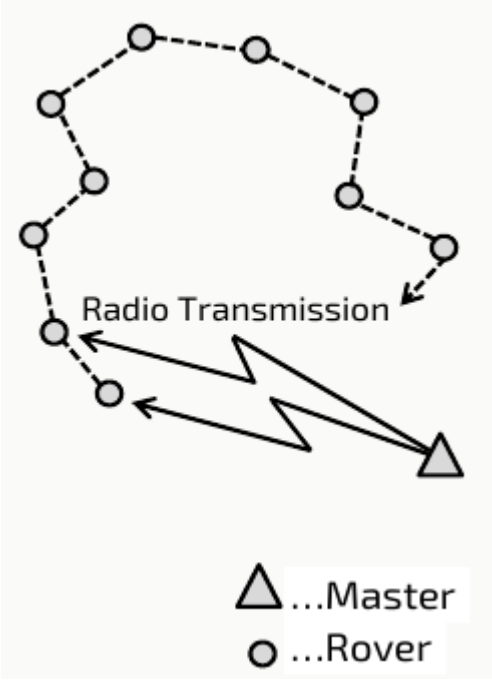
- Reference station with known position
- Observation errors (code) are determined at reference
- Corrections are transmitted to the rover (via radio or internet)
- Observation corrections using transmitted errors
- Accuracy: 0.5-5m, absolute
- Real-time: yes - Communication channel needed
- No reduction of multipath, non-line of sight errors



Real-Time Kinematic (RTK)

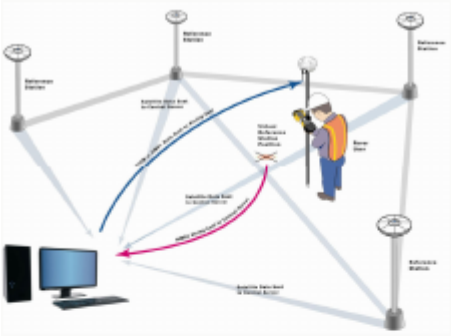
- **Multi-frequency code and carrier phase** observations at two receivers
- Carrier phase ambiguities need to be fixed!
- Master observations are transmitted to the rover via radio or internet
- Differencing at the rover eliminates many errors (real-time)
- Accuracy: 1-3m, relative (baseline length<10km)

- Storing master and rover data and processing them offline is called **PPK** (post-processing kinematic)



Network RTK

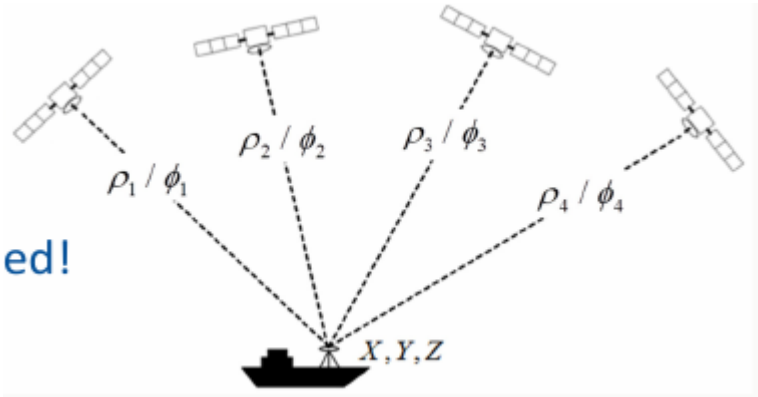
- Same as RTK, but master stations maintained by **reference network service** provider
- Network of master stations used to simulate master station close to rover
- Connection to network! Mostly via internet

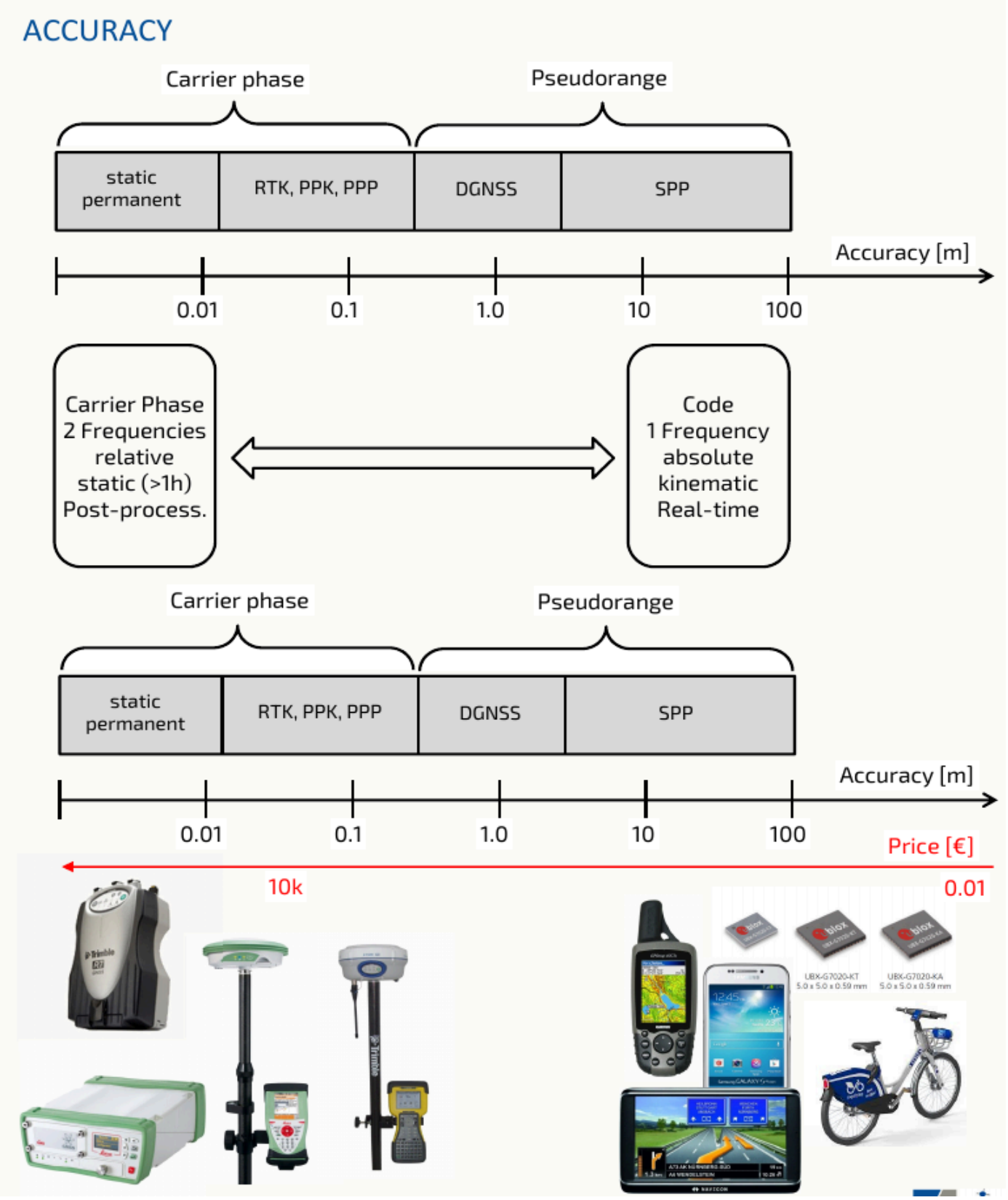


Precise Point Positioning

- Absolute positioning using code and carrier phase observations

- Error reduction via **accurate modelling** (not differencing)
- Phase ambiguities need to be fixed
- Model parameters via web-services
- Accuracy: ~cm
- Long conversion time (~30min)





- Output of GNSS processing is a 3D cartesian WGS84/ITRF coordinate